



Nanodiamonds @ N, P co-modified mesoporous carbon supported on macroscopic SiC foam for oxidative dehydrogenation of ethylbenzene

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ABSTRACT

A robust monolith catalyst has been fabricated involving the combination of nanodiamonds (NDs) and mesoporous carbon modified by nitrogen and phosphorus (ND@NMC-xP, x = amount of ammonium biphosphate) on a β -SiC foam. The ND@NMC-0.02 P/SiC monolith catalyst with 0.2 at.% of P elements exhibits an excellent catalytic performance for oxidative dehydrogenation of ethylbenzene (EB), delivering 90.2% selectivity to styrene (ST) up to 32.6% conversion. The specific conversion rate of monolith catalyst based on the unit weight of NDs and mesoporous carbons ($25.3 \text{ mmol g}_{\text{act. phase}}^{-1} \text{ h}^{-1}$) is 3-fold higher than that of unsupported NDs. The role of nitrogen is believed to enhance the density of active sites and thus offering a high EB conversion, whereas the addition of phosphorus inhibits deep oxidation of EB to carbon oxides and therefore for a high selectivity to ST. The monolith catalyst not only contributes a high utilization efficiency of NDs but also mitigates the high pressure drop across the catalytic bed, which is potentially suitable for a real industrial process.

1. Introduction

Styrene, as a very important chemical feedstock, is widely used to produce the rubber and peucine [1]. Typically, styrene synthesis in the industries comes from direct dehydrogenation of ethylbenzene using bulk iron-based catalysts. However, direct dehydrogenation reaction is an endothermic reaction and suffers the thermodynamic limitation which demands extremely high reaction temperature (i.e. 600 °C). In such a process, deactivation by coke is a main concern along with the heat management of this highly endothermic reaction. Therefore, a large amount of steam is used in order to mitigate coke deposition and reduce temperature loss [2–4]. The oxidative dehydrogenation (ODH) reaction of ethylbenzene catalyzed by nanocarbon, provides an attractive alternative process with obvious features of free coking and no equilibrium restrictions in ethylbenzene conversion [5–10]. Nanocarbons, such as carbon nanotubes (CNTs), graphene, nanodiamonds (NDs), are promising substituents for metal-based catalysts, given by their environmental benignity and controllable surface chemistry (electron density, acidity or basicity, etc) [11–18]. Among these

nanocarbon materials, NDs is a typical carbon-based dehydrogenation catalyst, which possesses small diameter and high surface activity than other carbon-based materials under dehydrogenation [3,19,20]. The ketonic carbonyl (C=O) groups on the carbon surface has been identified as the active sites for dehydrogenation reaction [21]. In addition, the surface structural defects can also be used as active sites for C–H bond activation [22,23]. Modification with heteroatoms enables tuning of the electronic properties of carbon-based catalysts by introducing electron acceptors or donors and consequently control the catalytic performance. Phosphorus and nitrogen atoms are often used to modify the electronic property of nanocarbon surfaces [24,25]. Heteroatom-modified nanodiamonds for the efficient alkane dehydrogenation has been reported recently [20,26,27]. It is reported that electron-rich nitrogen atoms can withdraw additional electrons from neighboring carbon atoms, thus improving the adsorption capacity of reactant during the reaction [22,25,28]. Modification with phosphate can effectively inhibit the over-oxidation of the product in oxidative dehydrogenation reaction and improve the selectivity and stability of the carbon-based nanomaterials. In addition, the incorporation of

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